GENERIC VALUE SEMIGROUPS WITH PRESCRIBED COORDINATE ORDERS

KNOWLEDGE SENSE

Abstract. Fix $d \geq 2$ and positive integers $r_1, \dots, r_d$, not necessarily distinct. We determine the value semigroup of the curve germ $t \mapsto (f_1(t), \dots, f_d(t))$ with $\operatorname{ord}_t f_i = r_i$ when the higher Taylor coefficients of the f_i are generic. List the weights $\alpha \cdot \mathbf{r} = \alpha_1 r_1 + \cdots + \alpha_d r_d$, $\alpha \in \mathbb{N}^d$, in nondecreasing order with multiplicity as $w_0 \leq w_1 \leq \cdots$; then the semigroup consists of the numbers $k_0 = 0$ and $k_j = \max(w_j, k_{j-1} + 1)$. Its genus is the maximum of $n - \#\{\alpha : \alpha \cdot \mathbf{r} < n\}$ over $n \geq 0$, and its conductor is the least n at which this maximum is attained. The proof combines a term-rank bound, which uses only the zero pattern of the matrix of Taylor coefficients, with the explicit germ $t \mapsto (t^{r_1} e^{\lambda_1 t}, \dots, t^{r_d} e^{\lambda_d t})$, whose relevant minors are positive by a total nonnegativity argument. We deduce that the genus differs from $\frac{d-1}{d}\big((d-1)!\, r_1 \cdots r_d\big)^{1/(d-1)}$ by at most $r_1 + \cdots + r_d + 1$. For $\mathbf{r} = (2\ell, 2\ell + 2, 2\ell + 4)$ the genus is $\frac{8}{3}\ell^{3/2} + O(\ell)$, so the lower bound of Cotterill, Kaplan and Vieira Costa has the correct leading term. For $\ell = 3$ the generic semigroup is $\langle 6, 8, 10, 17, 19, 21 \rangle$; the value 21 does not lie in the five-generator semigroup stated by Cotterill, Lima and Martins.

1. Introduction

Let $d \geq 2$ and let $\mathbf{r} = (r_1, \dots, r_d)$ be a vector of positive integers, not necessarily distinct. For formal power series

$$f_i(t) = t^{r_i}\big(1 + a_{i,1}t + a_{i,2}t^2 + \cdots\big) \in \mathbb{C}[[t]], \qquad 1 \leq i \leq d, \tag{1.1}$$

let $A_f \subseteq \mathbb{C}[[t]]$ be the image of $\mathbb{C}[[x_1, \dots, x_d]]$ under the substitution $x_i \mapsto f_i$, and let

$$S(f) = \{\operatorname{ord}_t h : 0 \neq h \in A_f\} \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$$

be its set of values. If the f_i are convergent, the image of the analytic algebra $\mathbb{C}\{x_1, \dots, x_d\}$ has the same set of values (Lemma 2.1). Rescaling the f_i does not change A_f, so we may assume that the leading coefficients are 1, as in (1.1). When $S(f)$ is a numerical semigroup, $t \mapsto (f_1(t), \dots, f_d(t))$ is a primitive parametrization of an irreducible curve germ, and the genus $\#(\mathbb{N} \setminus S(f))$ is its delta invariant; see [CLM22, Section 1.1] and [CKV23, Section 1.2].

This paper determines $S(f)$ when the coefficients $a_{i,k}$ are generic. The question arises in the study of Severi varieties of unicuspidal rational curves, where the semigroup of a cusp with generic higher coefficients plays a distinguished role [CLM22, CKV23]. There the cusp is described by its ramification profile, the sequence of distinct vanishing orders at $t = 0$ of linear combinations of the coordinate functions. If $r_1 < \cdots < r_d$, these orders are $r_1, \dots, r_d$, and the germs

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(1.1) with generic coefficients are the generic cusps with profile $\mathbf{r}$ in the sense of [CLM22, CKV23]. We also allow repeated entries, which give a different profile in their sense: for $\mathbf{r} = (2, 2)$ the difference $f_2 - f_1$ has order 3 for generic coefficients. For $d = 2$ the generic semigroup is known (Section 7.1). For general profiles, Cotterill, Kaplan and Vieira Costa remark that precisely determining it "appears to be difficult, if in principle algorithmically resolvable" [CKV23, Section 1.1]; for triples in arithmetic progression they bound its genus.

Enumerate $\mathbb{N}^d$ as $\alpha^{(0)} = 0, \alpha^{(1)}, \alpha^{(2)}, \dots$ so that the weights $w_j = \alpha^{(j)} \cdot \mathbf{r}$ are nondecreasing. Thus $w_0 \le w_1 \le \cdots$ is the list of all weights $\alpha \cdot \mathbf{r}$ with multiplicity; it does not depend on how ties are broken. Put

$$k_0 = 0, \qquad k_j = \max(w_j,\, k_{j-1} + 1) \quad (j \ge 1), \qquad G(\mathbf{r}) = \{k_j : j \ge 0\}. \tag{1.2}$$

Thus each weight is placed at the least integer that is at least the weight and not yet used; we call (k_j) the greedy sequence of $\mathbf{r}$. Equal entries of $\mathbf{r}$ still correspond to distinct coordinates, and their monomials are counted separately: for $\mathbf{r} = (2, 2)$ the weights are $0, 2, 2, 4, 4, 4, \dots$ and $G(\mathbf{r}) = \langle 2, 3\rangle$.

Theorem 1.1. *Let $d \ge 2$ and $\mathbf{r} \in \mathbb{Z}^d_{>0}$. Then $G(\mathbf{r})$ is a numerical semigroup. Let c be its conductor and $N = c + \min \mathbf{r}$. There exist polynomials $\Delta_1, \dots, \Delta_K$ with integer coefficients in the variables $a_{i,k}$ $(1 \le i \le d,\ 1 \le k < N)$, depending only on $\mathbf{r}$ and none of them identically zero, with the following property. If f is as in* (1.1) *and $\Delta_1(a) \cdots \Delta_K(a) \ne 0$, then*

$$S(f) = G(\mathbf{r})$$

for all values of the coefficients $a_{i,k}$ with $k \ge N$. The same holds for convergent f and the analytic algebra $\mathbb{C}\{f_1, \dots, f_d\}$. In particular $S(f) = G(\mathbf{r})$ whenever the coefficients $a_{i,k}$ are algebraically independent over $\mathbb{Q}$.

Example 1.2. For $\mathbf{r} = (6, 8, 10)$ the weights below 24, with multiplicity, are

$$0,\ 6,\ 8,\ 10,\ 12,\ 14,\ 16,\ 16,\ 18,\ 18,\ 20,\ 20,\ 22,\ 22,$$

and the recursion (1.2) gives the elements $0, 6, 8, 10, 12, 14$ and $16, 17, \dots, 23$ of $G(\mathbf{r})$ below 24. Since $G(\mathbf{r})$ is closed under addition (Theorem 1.1) and contains 6 and $16, \dots, 21$, it contains every integer $n \ge 16$. Hence

$$G(6, 8, 10) = \langle 6, 8, 10, 17, 19, 21\rangle,$$

with genus 10 and conductor 16. The ties $16 = 6{+}10 = 8{+}8$, $18 = 6{+}6{+}6 = 8{+}10$ and $20 = 6 + 6 + 8 = 10 + 10$ produce the values 17, 19 and 21; see Section 7.3.

For $n \ge 0$ put

$$M_{\mathbf{r}}(n) = \#\{\alpha \in \mathbb{N}^d : \alpha \cdot \mathbf{r} < n\}, \qquad F_{\mathbf{r}}(n) = n - M_{\mathbf{r}}(n).$$

Theorem 1.3. *Let $d \ge 2$, $\mathbf{r} \in \mathbb{Z}^d_{>0}$, and let g and c be the genus and the conductor of $G(\mathbf{r})$. Then*

$$g = \max_{n \ge 0} F_{\mathbf{r}}(n) = \max_{j \ge 0}(w_j - j), \qquad c = \min\{n \ge 0 : F_{\mathbf{r}}(n) = g\}.$$

If $a \le b$ are the two smallest entries of $\mathbf{r}$, counted with multiplicity, then $F_{\mathbf{r}}(n) \le 0$ for $n \ge 4ab$, so the maximum and the minimum above are attained below $4ab$. Moreover $G(\mathbf{r}) = \mathbb{N}$ if and only if $\min \mathbf{r} = 1$; if $m = \min \mathbf{r} \ge 2$, then $g \ge m - 1$ and every minimal generator of $G(\mathbf{r})$ is smaller than $c + m$.

Since $M_{\mathbf{r}}(n)$ counts lattice points in a simplex, volume estimates give the following bounds.

Theorem 1.4. *Let $d \geq 2$, $\mathbf{r} \in \mathbb{Z}^d_{>0}$, $P = r_1 \cdots r_d$, $R = r_1 + \cdots + r_d$ and $n_* = \big((d-1)!\,P\big)^{1/(d-1)}$. The genus g of $G(\mathbf{r})$ satisfies*

$$\frac{d-1}{d}\, n_* - R - 1 \;\leq\; g \;\leq\; \frac{d-1}{d}\, n_*.$$

If d is fixed and $r_i = \kappa_i \tau + O(1)$ with fixed $\kappa_i > 0$ as $\tau \to \infty$, then n_* has order $\tau^{d/(d-1)}$ and the error term is $O(\tau)$. For triples in arithmetic progression we obtain the following.

Corollary 1.5. *Let $m, \ell \geq 1$, $\mathbf{r} = (m\ell, m\ell + m, m\ell + 2m)$, and let g be the genus of $G(\mathbf{r})$. Then*

$$-3m(\ell+1) - 1 \;\leq\; g - \frac{(2m)^{3/2}}{3}\,\ell^{3/2} \;\leq\; \frac{2(2m)^{3/2}}{3}\,\ell^{1/2}.$$

In particular, for $\mathbf{r} = (2\ell, 2\ell+2, 2\ell+4)$ we have $g = \frac{8}{3}\ell^{3/2} + O(\ell)$.

Relation to earlier work. For $d = 2$ the generic value semigroup is that of a generic plane branch, generated by a, b and $ab/e + 1$, where $a \leq b$ are the orders and $e = \gcd(a, b)$; see [GMO26, Section 1 and Section 1.1] and Section 7.1. For $d = 3$, Cotterill, Kaplan and Vieira Costa bound the genus of the generic semigroup and give first approximations to it for arithmetic and supersymmetric triples [CKV23, Sections 2 and 3], and Cotterill, Lima and Martins state an explicit five-generator description for $(2m, 2m+2, 2m+4)$ [CLM22, Theorem 3.16]. Theorem 1.1 determines the generic semigroup uniformly for all $d \geq 2$ and all positive orders.

For $\mathbf{r} = (2\ell, 2\ell+2, 2\ell+4)$, Cotterill, Kaplan and Vieira Costa proved that $g > (\frac{8}{3} - \epsilon)\ell^{3/2}$ for large ℓ [CKV23, Corollary 2.5] and $g \leq \lceil \ell^2/2 \rceil + 2\ell$ [CKV23, Theorem 2.9], and they expected the lower bound to be closer to the truth. Corollary 1.5 confirms this: the lower bound has the correct leading term, and the same holds for their bound [CKV23, Corollary 2.7] for $(m\ell, m\ell + m, m\ell + 2m)$.

For $m = 3$ the semigroup in [CLM22, Theorem 3.16] is $\langle 6, 8, 10, 17, 19 \rangle$. It does not contain 21, which is a generic value by Example 1.2. Already the binomial $f_3^2 - f_1^2 f_2$ has order 21 for generic coefficients (Proposition 7.1). The same five-generator semigroups appear in [CKV23, Theorem 2.9] only as subsemigroups of the generic semigroup, which is consistent with Theorem 1.1.

Method. Modulo t^N, the algebra A_f is spanned by the finitely many monomials f^α with $\alpha \cdot \mathbf{r} < N$, and the values below N can be read off from the ranks of the submatrices formed by the first L rows of the matrix of their Taylor coefficients, $0 \leq L \leq N$ (Lemma 2.2). The column of f^α vanishes in the rows $0, \ldots, \alpha \cdot \mathbf{r} - 1$. Using only this zero pattern, a term-rank count bounds the rank of the first L rows by $\#\{j : k_j < L\}$ (Lemma 2.4). For the reverse inequality we show that suitable minors are nonzero polynomials in the coefficients by evaluating them at the germ $f_i = t^{r_i} e^{\lambda_i t}$. Then $f^\alpha = t^{\alpha \cdot \mathbf{r}} e^{(\alpha \cdot \lambda) t}$, so the exponents $\alpha \cdot \lambda$ are determined by $\lambda_1, \ldots, \lambda_d$ and cannot be chosen independently. Choosing $\lambda_i = C r_i + \omega_i$ makes them increase with the weight. Each column of the resulting matrix is a column of $\exp(\lambda \Lambda)$ for some $\lambda \geq 0$, where Λ is the shift matrix. The matrices $\exp(\lambda\Lambda)$ are totally nonnegative, and the minors we need are positive because one term of a Cauchy–Binet expansion is positive (Lemma 3.2). Their nonvanishing is also a

statement about poised Hermite–Birkhoff interpolation problems and follows from a theorem of Atkinson and Sharma [AS69] (Remark 3.3). Apart from the comparison in Section 7.1, the proofs are self-contained.

Organization. Section 2 contains the combinatorics of the sequence (k_j) and the reduction to jets. Section 3 proves the positivity lemma. Section 4 proves Theorem 1.1, Section 5 proves Theorem 1.3 and describes the computation, and Section 6 proves Theorem 1.4 and Corollary 1.5. Section 7 compares the results with [GMO26], [CKV23] and [CLM22].

2. Jets and the greedy sequence

For $h \in \mathbb{C}[[t]]$ and $N \geq 1$ let $j_N(h) \in \mathbb{C}^N$ be the vector of the coefficients of $t^0, \dots, t^{N-1}$. For $0 \neq v \in \mathbb{C}^N$ let $\operatorname{ord} v$ be the least index of a nonzero coordinate.

Lemma 2.1. *Let f be as in (1.1) and $N \geq 1$. Then*

$$\{j_N(h) : h \in A_f\} = V_N(f) := \operatorname{span}\{j_N(f^\alpha) : \alpha \cdot \mathbf{r} < N\},$$

and $S(f) \cap [0, N) = \{\operatorname{ord} v : 0 \neq v \in V_N(f)\}$. The same holds with A_f replaced by the polynomial algebra $\mathbb{C}[f_1, \dots, f_d]$, or, for convergent f, by $\mathbb{C}\{f_1, \dots, f_d\}$.

Proof. Since $\operatorname{ord} f^\alpha = \alpha \cdot \mathbf{r}$, every $P = \sum_\alpha c_\alpha x^\alpha$ satisfies

$$j_N(P(f)) = \sum_{\alpha \cdot \mathbf{r} < N} c_\alpha \, j_N(f^\alpha);$$

conversely every vector of $V_N(f)$ is the jet of a polynomial in the f_i. For $h \neq 0$, $\operatorname{ord} h < N$ holds if and only if $j_N(h) \neq 0$, and then $\operatorname{ord} h = \operatorname{ord} j_N(h)$. □

For a subspace $V \subseteq \mathbb{C}^N$ and $0 \leq L \leq N$ let $\pi_L : \mathbb{C}^N \to \mathbb{C}^L$ be the projection onto the first L coordinates and $\rho_V(L) = \dim \pi_L(V)$.

Lemma 2.2. *Let $V \subseteq \mathbb{C}^N$ and $0 \leq L < N$. Then $\rho_V(L+1) - \rho_V(L) \in \{0, 1\}$, and $L = \operatorname{ord} v$ for some $0 \neq v \in V$ if and only if $\rho_V(L+1) = \rho_V(L) + 1$.*

Proof. Let $K_L = V \cap \ker \pi_L$, the set consisting of 0 and the vectors of order at least L. Then $\dim K_L = \dim V - \rho_V(L)$, and K_{L+1} is the kernel of the coordinate of index L on K_L, so it has codimension 0 or 1 in K_L. The codimension is 1 exactly when some vector of V has order L. □

Lemma 2.3. *Let $d \geq 2$ and let (w_j) and (k_j) be as in (1.2).*

(a) *$k_j = j + \max_{0 \leq i \leq j}(w_i - i)$. In particular (k_j) is strictly increasing and $k_j \geq w_j$.*
(b) *The maximum $D = \max_{j \geq 0}(w_j - j)$ exists. Let J be the least index with $w_J - J = D$. Then $k_j = j + D$ for $j \geq J$, and $k_{J-1} \leq J + D - 2$ if $J > 0$.*
(c) *$G(\mathbf{r})$ is cofinite in $\mathbb{N}$, with $\#(\mathbb{N} \setminus G(\mathbf{r})) = D$ and conductor $J + D = w_J$. Moreover $D = 0$ if and only if $J = 0$ if and only if $G(\mathbf{r}) = \mathbb{N}$.*

Proof. (a) By induction: $k_0 = 0 = w_0$, and $k_j = \max\big(w_j, \, j + \max_{i<j}(w_i - i)\big) = j + \max_{i \leq j}(w_i - i)$.

(b) Here we use $d \geq 2$. All α with $\alpha \cdot \mathbf{r} < w_j$ precede the index j. Counting only the vectors supported on the first two coordinates with $\alpha_1 r_1 < w_j/2$ and $\alpha_2 r_2 < w_j/2$, we get

$$j \geq M_{\mathbf{r}}(w_j) \geq \left\lceil \frac{w_j}{2r_1} \right\rceil \left\lceil \frac{w_j}{2r_2} \right\rceil \geq \frac{w_j^2}{4r_1 r_2}.$$

Hence $w_j - j \le w_j - w_j^2/(4r_1r_2)$. Since only finitely many α have a given weight, $w_j \to \infty$, so $w_j - j \to -\infty$ and the maximum D exists. For $j \ge J$ we get $k_j = j + D$ from (a). If $J > 0$, the minimality of J gives $w_i - i \le D - 1$ for $i < J$, so $k_{J-1} \le J - 1 + D - 1$.

(c) The elements of $G(\mathbf{r})$ below k_j are $k_0, \dots, k_{j-1}$, so exactly $k_j - j$ integers below k_j are missing from $G(\mathbf{r})$. For $j \ge J$ this number is D, and $G(\mathbf{r}) \supseteq \{J + D, J+D+1, \dots\}$. If $J > 0$, then $J+D-1$ lies strictly between k_{J-1} and $k_J = J+D$, so it is a gap and the conductor is $J + D = w_J$. If $J = 0$, then $D = w_0 = 0$ and $k_j = j$ for all j. □

The following bound uses only the positions of the structural zeros.

Lemma 2.4. *Let $p \ge 1$ and $L \ge 0$, and let Q be an $L \times p$ matrix, with rows indexed by $0, \dots, L-1$ and columns by $0, \dots, p-1$, such that $Q_{u,j} = 0$ whenever $u < w_j$. Then* $\operatorname{rank} Q \le \mu(L) := \#\{j < p : k_j < L\}$.

Proof. A nonzero minor of size s has a nonzero term in its Leibniz expansion, that is, s nonzero entries of Q in distinct rows and columns. It suffices to bound the size of such a set T of entries by $m = \mu(L)$. If $m = p$ there is nothing to prove, so let $m < p$; then $k_m \ge L$, and by Lemma 2.3(a) there is $i \le m$ with $m + w_i - i \ge L$. Every column $j \ge i$ satisfies $w_j \ge w_i \ge L - (m - i)$, so its nonzero entries lie in the at most $m - i$ rows $L - (m - i), \dots, L - 1$. Therefore T contains at most i entries in the columns $0, \dots, i - 1$ and at most $m - i$ entries in the other columns. □

The bound is attained by the pattern of positions (k_j, j), $j < \mu(L)$, and Proposition 4.1 shows that equality holds for the jet matrix of a generic germ.

3. A positivity lemma

A real matrix is *totally nonnegative* if all its minors are nonnegative. Fix $N \ge 1$, let Λ be the $N \times N$ lower shift matrix, $\Lambda e_u = e_{u+1}$ for $u < N - 1$ and $\Lambda e_{N-1} = 0$, and for $s \in \mathbb{R}$ put

$$E(s) = \sum_{k=0}^{N-1} \frac{s^k \Lambda^k}{k!}, \qquad E(s)_{u,v} = \begin{cases} s^{u-v}/(u-v)!, & u \ge v, \\ 0, & u < v, \end{cases}$$

with rows and columns indexed by $0, \dots, N-1$. Then $E(s)E(s') = E(s+s')$, and the column $E(s)e_v$ is the jet $j_N(t^v e^{st})$.

Lemma 3.1. *For $s \ge 0$ the matrix $E(s)$ is totally nonnegative.*

Proof. Let B be a lower bidiagonal matrix with nonnegative entries, and consider the submatrix on rows $i_1 < \cdots < i_p$ and columns $j_1 < \cdots < j_p$. If $j_1 > i_1$, its first row vanishes. If $j_1 < i_1 - 1$, its first column vanishes. If $j_1 = i_1$, the only possibly nonzero entry of its first row is the corner entry, and if $j_1 = i_1 - 1$ the same holds for its first column. By induction every minor of B is either zero or a product of entries of B, hence nonnegative. Products of totally nonnegative matrices are totally nonnegative by the Cauchy–Binet formula, so $(I + s\Lambda/n)^n$ is totally nonnegative for every $n \ge 1$. Since $\Lambda^N = 0$, we have $(I + s\Lambda/n)^n = \sum_{k<N} \binom{n}{k} (s/n)^k \Lambda^k \to E(s)$ as $n \to \infty$, and minors depend continuously on the entries. □

Lemma 3.2. *Let $q \ge 1$, let $0 < \lambda_1 < \cdots < \lambda_q$ be real numbers, and let $0 \le a_1 \le \cdots \le a_q < N$ be integers. Let X be the $N \times q$ matrix with columns $E(\lambda_j)e_{a_j}$. Then*

X is totally nonnegative, and for all integers $u_1 < \cdots < u_q < N$ with $a_j \le u_j$ for every j,

$$\det\Big(\frac{\lambda_j^{u_i-a_j}}{(u_i-a_j)!}\Big)_{1\le i,j\le q} > 0,$$

where an entry with $u_i < a_j$ is read as 0. This determinant is the minor of X on the rows $u_1,\dots,u_q$.

Proof. We use induction on q. For $q = 1$ the column X has nonnegative entries, and its entry in row $u_1 \ge a_1$ is $\lambda_1^{u_1-a_1}/(u_1-a_1)! > 0$.

Let $q \ge 2$. The $N \times (q-1)$ matrix Y with columns $E(\lambda_j-\lambda_1)e_{a_j}$, $2 \le j \le q$, satisfies the hypotheses with the nodes $0 < \lambda_2-\lambda_1 < \cdots < \lambda_q-\lambda_1$. By induction, Y is totally nonnegative and its minor on the rows $u_2,\dots,u_q$ is positive. Since $E(\lambda_j) = E(\lambda_1)E(\lambda_j-\lambda_1)$, we have $X = E(\lambda_1)B$ with $B = (e_{a_1} \mid Y)$.

We claim that B is totally nonnegative. Minors not involving the first column are minors of Y. Consider a minor of B on rows $t_1 < \cdots < t_p$ that involves the first column. If a_1 is not among the t_i, the first column of the submatrix vanishes. If $a_1 = t_s$ with $s \ge 2$, then the row $t_1 < a_1$ vanishes, because the column e_{a_1} is zero there and the column of Y indexed by j is zero in the rows of index less than $a_j \ge a_1$. If $a_1 = t_1$, the first column of the submatrix is $(1,0,\dots,0)^{\mathsf T}$, and the minor equals a minor of Y. In all cases the minor is nonnegative.

By the Cauchy–Binet formula X is totally nonnegative, and for $U = (u_1,\dots,u_q)$

$$\det X[U] = \sum_T \det E(\lambda_1)[U,T]\,\det B[T],$$

where T runs over the increasing q-tuples of row indices. All terms are nonnegative. Take $T_0 = (a_1,u_2,\dots,u_q)$, which is increasing because $a_1 \le u_1 < u_2$. By the previous paragraph, $\det B[T_0] = \det Y[(u_2,\dots,u_q)] > 0$. The matrix $E(\lambda_1)[U,T_0]$ is lower triangular: for $j > i$ its (i,j) entry is $E(\lambda_1)_{u_i,u_j} = 0$, because $u_i < u_j$. Its diagonal is $\lambda_1^{u_1-a_1}/(u_1-a_1)!, 1, \dots, 1$, which is positive. Hence the term for T_0 is positive, and so is $\det X[U]$. □

Remark 3.3. The nonvanishing part of Lemma 3.2 is a classical statement about Hermite–Birkhoff interpolation. Put $n_0 = u_q$ and consider polynomials of degree at most n_0 in the basis $x^h/h!$, $0 \le h \le n_0$. Prescribe $\phi^{(h)}(0)$ for $h \in \{0,\dots,n_0\} \setminus \{u_1,\dots,u_q\}$ and $\phi^{(a_j)}(\lambda_j)$ for $1 \le j \le q$; these are n_0+1 conditions. The condition at 0 of order h picks out the coefficient of $x^h/h!$, and the condition of order a at λ sends $x^u/u!$ to $\lambda^{u-a}/(u-a)!$. Expanding the determinant of the problem along the conditions at 0 leaves, up to sign and transposition, the determinant of Lemma 3.2. The number of conditions of order at most p is

$$(p+1) - \#\{j : u_j \le p\} + \#\{j : a_j \le p\} \ge p+1,$$

because $a_j \le u_j$; this is the Pólya condition. Every sequence at the node 0 is conservative in the sense of [AS69, Definition 2], since 0 is the leftmost node. The node λ_j carries a single condition, of order a_j, and every node carrying a condition of order smaller than a_j is 0 or some λ_i with $a_i < a_j$, hence $i < j$ and $\lambda_i < \lambda_j$; so this sequence is conservative too. By [AS69, Theorem 2] the problem is poised, and the determinant is nonzero. The proof above is self-contained and also gives the sign.

4. Proof of Theorem 1.1

Fix $N \geq 1$. Let $\Omega_N = \{\alpha \in \mathbb{N}^d : \alpha \cdot \mathbf{r} < N\}$, put $p = \#\Omega_N = M_{\mathbf{r}}(N)$ and $\omega = (1, N, N^2, \ldots, N^{d-1})$. For $\alpha \in \Omega_N$ every coordinate satisfies $\alpha_i \leq N-1$, so the numbers $\alpha \cdot \omega$ are pairwise distinct. Enumerate $\Omega_N = \{\alpha^{(0)}, \ldots, \alpha^{(p-1)}\}$ by increasing weight, and among vectors of equal weight by increasing $\alpha \cdot \omega$. This is an initial segment of an enumeration as in (1.2), so the corresponding w_j and k_j, $j < p$, are those of (1.2).

Let $X_N(f)$ be the $N \times p$ matrix with columns $j_N(f^{\alpha^{(j)}})$. Its entries are polynomials with integer coefficients in the $a_{i,k}$, $1 \leq k < N$, and $X_N(f)_{u,j} = 0$ for $u < w_j$. Let $m_N = \#\{j : k_j < N\}$; note that $k_j < N$ implies $w_j < N$, hence $j < p$. For $1 \leq q \leq m_N$ let $\Delta_q^{(N)}$ be the minor of $X_N(f)$ on the rows $k_0, \ldots, k_{q-1}$ and the columns $0, \ldots, q-1$.

Proposition 4.1. *If $\Delta_q^{(N)}(a) \neq 0$ for $1 \leq q \leq m_N$, then*

$$S(f) \cap [0, N) = G(\mathbf{r}) \cap [0, N).$$

Proof. Let $0 \leq L \leq N$. The column space of the $L \times p$ submatrix of $X_N(f)$ formed by its first L rows is $\pi_L(V_N(f))$, so this submatrix has rank $\rho(L) = \rho_{V_N(f)}(L)$. Lemma 2.4 gives $\rho(L) \leq \mu(L) = \#\{j < p : k_j < L\}$; since $k_j < L \leq N$ implies $w_j < N$ and hence $j < p$, also $\mu(L) = \#\{j : k_j < L\}$. If $\mu(L) = q \geq 1$, the rows $k_0, \ldots, k_{q-1}$ are smaller than L, and the nonzero minor $\Delta_q^{(N)}$ shows $\rho(L) \geq q$. Hence $\rho(L) = \mu(L)$ for all $L \leq N$. By Lemmas 2.1 and 2.2, an integer $L < N$ lies in $S(f)$ if and only if $\mu(L+1) = \mu(L) + 1$, that is, if and only if $L \in G(\mathbf{r})$. □

Proposition 4.2. *None of the polynomials $\Delta_q^{(N)}$, $1 \leq q \leq m_N$, is identically zero. More precisely, let $C = 1 + \max_{\alpha \in \Omega_N} \alpha \cdot \omega$, $\lambda_i = Cr_i + \omega_i$, and $f_i = t^{r_i} e^{\lambda_i t}$, that is, $a_{i,k} = \lambda_i^k / k!$. Then $\Delta_q^{(N)}(a) > 0$ for $1 \leq q \leq m_N$.*

Proof. For $\alpha \in \Omega_N$ we have $f^\alpha = t^{\alpha \cdot \mathbf{r}} e^{\lambda_\alpha t}$ with

$$\lambda_\alpha = \alpha \cdot \lambda = C\,(\alpha \cdot \mathbf{r}) + \alpha \cdot \omega,$$

so the j-th column of $X_N(f)$ is $E(\lambda_{(j)}) e_{w_j}$, where $\lambda_{(j)} = \lambda_{\alpha^{(j)}}$. The numbers $\lambda_{(j)}$ are strictly increasing in j. Indeed, if $w_j < w_{j'}$, then $\lambda_{(j)} \leq Cw_j + C - 1 < Cw_{j'} \leq \lambda_{(j')}$; if $w_j = w_{j'}$ and $j < j'$, the tie-breaking rule gives $\alpha^{(j)} \cdot \omega < \alpha^{(j')} \cdot \omega$. Moreover $\lambda_{(0)} = 0$ and $\lambda_{(j)} > 0$ for $j \geq 1$.

The column 0 is $E(0)e_0 = e_0$, and every column $j \geq 1$ has weight $w_j \geq \min \mathbf{r} \geq 1$, hence vanishes in row $0 = k_0$. Expanding along this row, $\Delta_q^{(N)}$ equals the minor on the rows $k_1, \ldots, k_{q-1}$ and the columns $1, \ldots, q-1$ (and $\Delta_1^{(N)} = 1$). This is the minor of Lemma 3.2 with $q-1$ columns, nodes $\lambda_{(1)} < \cdots < \lambda_{(q-1)}$, orders $a_j = w_j$ and rows $u_j = k_j \geq w_j$; it is positive. □

Proof of Theorem 1.1. First, $G(\mathbf{r})$ is a numerical semigroup. It contains 0 and is cofinite by Lemma 2.3(c). Let $x, y \in G(\mathbf{r})$ and $N > x + y$. By Proposition 4.2 there is f with $\Delta_1^{(N)} \cdots \Delta_{m_N}^{(N)}(a) \neq 0$, and by Proposition 4.1 there are $h_x, h_y \in A_f$ of orders x and y. Then $h_x h_y \in A_f$ has order $x + y < N$, so $x + y \in S(f) \cap [0, N) = G(\mathbf{r}) \cap [0, N)$.

Now let c be the conductor of $G(\mathbf{r})$, let $m = \min \mathbf{r} = r_{i_0}$, and take $N = c + m$ and $\Delta_\nu = \Delta_\nu^{(N)}$, $1 \leq \nu \leq K = m_N$. These polynomials depend only on $\mathbf{r}$, are not identically zero by Proposition 4.2, and involve only the $a_{i,k}$ with $k < N$. Suppose

that $\Delta_1(a)\cdots\Delta_K(a)\neq 0$. By Proposition 4.1, $S(f)\cap[0,c+m)=G(\mathbf{r})\cap[0,c+m)$. In particular A_f contains elements $h_c,\dots,h_{c+m-1}$ of the orders $c,\dots,c+m-1$, and the products $h_{c+\nu}f_{i_0}^k$ show that every integer $n\geq c$ lies in $S(f)$. Since also $S(f)\cap[0,c)=G(\mathbf{r})\cap[0,c)$, we get $S(f)=G(\mathbf{r})$. The argument used only jets, so by Lemma 2.1 it applies verbatim to $\mathbb{C}\{f_1,\dots,f_d\}$ for convergent f. Finally, a nonzero polynomial with rational coefficients does not vanish at a point whose coordinates are algebraically independent over $\mathbb{Q}$. □

Remark 4.3. (a) For $N=c+\min\mathbf{r}$, Proposition 4.2 makes all the minors $\Delta_q^{(N)}$ nonzero at one point, so the entire germ $t\mapsto(t^{r_1}e^{\lambda_1 t},\dots,t^{r_d}e^{\lambda_d t})$ constructed there has value semigroup exactly $G(\mathbf{r})$.

(b) Only the coefficients $a_{i,k}$ with $k<N$ enter the condition of Theorem 1.1. Hence $S(f)=G(\mathbf{r})$ also when each f_i is a polynomial $t^{r_i}(1+a_{i,1}t+\cdots+a_{i,D_i}t^{D_i})$ with $D_i\geq N-1$ and generic coefficients. Germs with $\Delta_1(a)\cdots\Delta_K(a)=0$ may have other value semigroups, and Theorem 1.1 does not describe them.

(c) We work over $\mathbb{C}$. The polynomials Δ_ν are not identically zero and have integer coefficients, but we make no claim in positive characteristic. For $d=1$ the values are the multiples of r_1, which form a numerical semigroup only when $r_1=1$.

5. Genus, conductor and computation

Proof of Theorem 1.3. Let D and J be as in Lemma 2.3, so that $g=D=\max_j(w_j-j)$ and $c=w_J$ if $D>0$, $c=0$ if $D=0$.

For every $n\geq 0$, the elements of $G(\mathbf{r})$ below n are the $k_j<n$, and $k_j\geq w_j$, so there are at most $\#\{j:w_j<n\}=M_{\mathbf{r}}(n)$ of them. Thus at least $F_{\mathbf{r}}(n)$ gaps lie below n, and $F_{\mathbf{r}}(n)\leq g$.

If $D=0$, then $F_{\mathbf{r}}(0)=0=g$ and $c=0$. Let $D>0$. If $i<J$ had $w_i=w_J$, then $w_i-i>w_J-J=D$, which is impossible; hence the indices j with $w_j<w_J$ are exactly $0,\dots,J-1$, $M_{\mathbf{r}}(w_J)=J$ and $F_{\mathbf{r}}(c)=w_J-J=D$. If $n<c$, the gap $c-1$ is not below n, so at most $g-1$ gaps lie below n and $F_{\mathbf{r}}(n)\leq g-1$. This proves the formulas for g and c.

Let $n\geq 4ab$, and let $i_1\neq i_2$ be positions with $r_{i_1}=a$ and $r_{i_2}=b$. The vectors supported on $\{i_1,i_2\}$ with $\alpha_{i_1}a<n/2$ and $\alpha_{i_2}b<n/2$ have weight less than n, so

$$M_{\mathbf{r}}(n)\geq\left\lceil\frac{n}{2a}\right\rceil\left\lceil\frac{n}{2b}\right\rceil\geq\frac{n^2}{4ab}\geq n,$$

and $F_{\mathbf{r}}(n)\leq 0$. If $m=\min\mathbf{r}=1$, then the multiples of the corresponding coordinate give $M_{\mathbf{r}}(n)\geq n$ for all n, so $g=0$ and $G(\mathbf{r})=\mathbb{N}$. If $m\geq 2$, only $\alpha=0$ has weight less than m, so $g\geq F_{\mathbf{r}}(m)=m-1>0$. Finally, an element $n\geq c+m$ of $G(\mathbf{r})$ is the sum of $m\in G(\mathbf{r})$ and $n-m\geq c$, so it is not a minimal generator. □

Remark 5.1. Let $\varpi(n)=\#\{\alpha:\alpha\cdot\mathbf{r}=n\}$, so that

$$\sum_{n\geq 0}\varpi(n)z^n=\prod_{i=1}^{d}\frac{1}{1-z^{r_i}}.$$

The numbers $\varpi(n)$, $n<n_1$, are obtained by the usual recursion for unbounded knapsack counts, one pass per coordinate (also for repeated orders), with $O(dn_1)$ additions. Put $\beta_n=\#\{j:w_j\leq n\leq k_j\}$, the number of indices whose weight has

been reached but which have not been placed below n. If $\beta_n > 0$, the least such j satisfies $k_j = n$; hence $n \in G(\mathbf{r})$ if and only if $\beta_n > 0$, and

$$\beta_0 = 1, \qquad \beta_{n+1} = \beta_n - [\beta_n > 0] + \varpi(n+1).$$

By Theorem 1.3 it suffices to take $n_1 = 4ab + \min \mathbf{r}$, which yields the gaps, g, c and the minimal generators in $O(dab)$ additions of integers. The cost is polynomial in the orders, but not in their bit length.

Table 1 lists some values obtained in this way. The row $(4, 6)$ agrees with the plane case of Section 7.1, and the row $(6, 8, 10)$ is Example 1.2. The values in Tables 1 and 2 and in Section 7.3 were computed with this procedure; a short program carrying out these computations is distributed with the paper.

TABLE 1. Generic value semigroups for some coordinate orders.

$\mathbf{r}$	genus	conductor	minimal generators
$(2, 2)$	1	2	$2, 3$
$(4, 6)$	8	16	$4, 6, 13$
$(4, 4, 4)$	4	8	$4, 5, 6$
$(3, 3, 3, 3)$	2	3	$3, 4, 5$
$(5, 7, 9)$	8	14	$5, 7, 9$
$(6, 8, 10)$	10	16	$6, 8, 10, 17, 19, 21$
$(8, 10, 12)$	16	28	$8, 10, 12, 21, 25$
$(10, 12, 14)$	22	34	$10, 12, 14, 25, 41, 43$
$(12, 15, 20)$	35	59	$12, 15, 20, 61$
$(20, 22, 24)$	67	104	$20, 22, 24, 45, 95, 121, 123$

6. Asymptotics

Lemma 6.1. *Let $P = r_1 \cdots r_d$ and $R = r_1 + \cdots + r_d$. For every $n \geq 0$,*

$$\frac{n^d}{d!\,P} \leq M_{\mathbf{r}}(n) \leq \frac{(n+R)^d}{d!\,P}.$$

Proof. The cubes $\alpha + [0,1)^d$, $\alpha \in \mathbb{N}^d$, are disjoint. If $x \in \mathbb{R}^d_{\geq 0}$ and $x \cdot \mathbf{r} < n$, then $\alpha = \lfloor x \rfloor$ satisfies $\alpha \cdot \mathbf{r} \leq x \cdot \mathbf{r} < n$; hence the simplex $\{x \geq 0 : x \cdot \mathbf{r} < n\}$, of volume $n^d/(d!P)$, is covered by the cubes with $\alpha \cdot \mathbf{r} < n$. Conversely, each such cube lies in $\{x \geq 0 : x \cdot \mathbf{r} < n + R\}$, of volume $(n+R)^d/(d!P)$. □

Proof of Theorem 1.4. Put $\phi(x) = x - x^d/(d!P)$ for $x \geq 0$. It is concave, with $\phi'(x) = 1 - x^{d-1}/((d-1)!P) \leq 1$, and it attains its maximum at n_*, where

$$\phi(n_*) = n_* - n_* \cdot \frac{n_*^{d-1}}{d!\,P} = n_* - \frac{n_*}{d} = \frac{d-1}{d}\, n_*.$$

By Lemma 6.1, $F_{\mathbf{r}}(n) \leq \phi(n) \leq \phi(n_*)$, and Theorem 1.3 gives the upper bound.

If $n_* < R$, then $\frac{d-1}{d} n_* - R - 1 < 0 \leq g$. Let $n_* \geq R$, put $n = \lfloor n_* - R \rfloor \geq 0$ and $x = n + R \in (n_* - 1, n_*]$; note $x \geq 0$. By Lemma 6.1,

$$F_{\mathbf{r}}(n) \geq n - \frac{(n+R)^d}{d!\,P} = \phi(x) - R.$$

Since $0 \leq \phi' \leq 1$ on $[x, n_*]$, we have $\phi(x) \geq \phi(n_*) - (n_* - x) \geq \phi(n_*) - 1$. Hence $g \geq F_{\mathbf{r}}(n) \geq \frac{d-1}{d} n_* - R - 1$. □

Proof of Corollary 1.5. Here $d = 3$, $P = m^3\ell(\ell+1)(\ell+2)$, $R = 3m(\ell+1)$ and $n_* = (2P)^{1/2}$, so $\frac{2}{3}n_* = \frac{(2m)^{3/2}}{3}\sigma$ with $\sigma = (\ell(\ell+1)(\ell+2))^{1/2}$. Using $(1+x)^{1/2} \le 1 + x/2$,

$$\ell^{3/2} \le \sigma \le \ell^{3/2}\Big(1+\frac{1}{2\ell}\Big)\Big(1+\frac{1}{\ell}\Big) \le \ell^{3/2} + \frac{3}{2}\ell^{1/2} + \frac{1}{2}\ell^{-1/2} \le \ell^{3/2} + 2\ell^{1/2}.$$

By Theorem 1.4,

$$g \le \frac{(2m)^{3/2}}{3}\,\sigma \le \frac{(2m)^{3/2}}{3}\,\ell^{3/2} + \frac{2(2m)^{3/2}}{3}\,\ell^{1/2},$$
$$g \ge \frac{(2m)^{3/2}}{3}\,\sigma - R - 1 \ge \frac{(2m)^{3/2}}{3}\,\ell^{3/2} - 3m(\ell+1) - 1.$$

For $m = 2$ the constant is $\frac{(2m)^{3/2}}{3} = \frac{8}{3}$. □

Table 2 compares the exact genus with the bounds of Theorem 1.4 and with the upper bound $\lceil \ell^2/2 \rceil + 2\ell$ of [CKV23, Theorem 2.9], for $\mathbf{r} = (2\ell, 2\ell+2, 2\ell+4)$. The lower bound is rounded up and the upper bound is rounded down.

TABLE 2. The genus g of $G(2\ell, 2\ell+2, 2\ell+4)$.

ℓ	lower bound	g	upper bound	$g/\ell^{3/2}$	$\lceil \ell^2/2 \rceil + 2\ell$
10	30	67	96	2.119	70
30	274	382	460	2.325	510
100	2100	2472	2706	2.472	5200
300	12119	13262	13925	2.552	45600
1000	78447	82338	84453	2.604	502000

7. Comparison with earlier results

7.1. **Two coordinates.** Let $d = 2$, $\mathbf{r} = (a, b)$ with $a \le b$, and $e = \gcd(a, b)$. Reparametrizing by $s = t\,(f_1/t^a)^{1/a}$ we may write $f_1 = s^a$ and $f_2 = s^b(1 + \lambda s + \cdots)$ with $\lambda = a_{2,1} - \frac{b}{a}\,a_{1,1}$. For $\lambda \neq 0$ these are the generic plane curve singularities studied by Gorsky, Mazin and Oblomkov, whose semigroup is generated by a, b and $ab/e + 1$ [GMO26, Section 1 and Section 1.1]; the delta invariant is $\big((a-1)(b-1) + e - 1\big)/2$, which is the box count in [GMO26, Theorem 1.1(b)]. Let N be as in Theorem 1.1 and $N' = \max(N, 2)$. The conditions $\lambda \neq 0$ and $\Delta_1(a) \cdots \Delta_K(a) \neq 0$ define nonempty Zariski open subsets of the irreducible affine space of coefficients $a_{i,k}$, $k < N'$, so they hold simultaneously for generic coefficients. Comparing the two descriptions gives

$$G(a, b) = \langle a,\, b,\, ab/e + 1 \rangle.$$

This identity is not evident from (1.2); here it follows by comparing two descriptions of the generic semigroup. For coprime orders the third generator is redundant and $G(a, b) = \langle a, b \rangle$, and for $a = b$ one gets $\langle a, a+1 \rangle$.

7.2. **Arithmetic and supersymmetric triples.** Cotterill, Kaplan and Vieira Costa [CKV23] call the value semigroup of a germ with prescribed orders and algebraically independent higher coefficients the generic semigroup adapted to the profile; there $\mathbf{r}$ is strictly increasing, and for such $\mathbf{r}$ this semigroup is $G(\mathbf{r})$ by Theorem 1.1. For $\mathbf{r} = (2\ell, 2\ell+2, 2\ell+4)$ they prove $g > (\frac{8}{3} - \epsilon)\ell^{3/2}$ for large ℓ [CKV23, Corollary 2.5], a containment $S^* \subseteq S$ for an explicit five-generator semigroup S^* together with $g \leq \lceil \ell^2/2 \rceil + 2\ell$ [CKV23, Theorem 2.9], and the analogous statements for $(m\ell, m\ell+m, m\ell+2m)$ [CKV23, Corollary 2.7, Theorem 2.10], the latter under the assumptions $m \geq 2$ and $\ell \geq 2m$. After [CKV23, Theorem 2.9] they remark that the lower bound seems closer to the true genus than the upper bound. Corollary 1.5 shows that the lower bounds of [CKV23, Corollaries 2.5 and 2.7] have the correct leading terms and that the genus has order $\ell^{3/2}$; Table 2 illustrates this. The containment $S^* \subseteq S$ is consistent with Theorem 1.1; for $\ell = 3$ it is strict (Section 7.3).

In [CKV23, Remark 2.8] the generic semigroup is conjectured to be the limit of a discrete dynamical system that inserts new elements at Betti elements. Theorem 1.1 determines the limit object, but we do not prove that it is produced by that procedure.

For supersymmetric triples $\mathbf{r} = (ab, ac, bc)$ with pairwise coprime $2 \leq a < b < c$, [CKV23, Section 3] studies the generic semigroup through the first approximation $S' = \langle ab, ac, bc, abc+1 \rangle$. When $abc + 1 \notin \langle ab, ac, bc \rangle$, it computes the genus of S', which is an upper bound for the genus of the generic semigroup [CKV23, Proposition 3.12]. For $(a, b, c) = (3, 4, 5)$ Theorem 1.1 gives $G(12, 15, 20) = \langle 12, 15, 20, 61 \rangle = S'$, so the generic genus 35 attains the upper bound of [CKV23, Proposition 3.12].

7.3. **The five-generator statement for consecutive even orders.** We use the numbering of the arXiv version of [CLM22]. Its Theorem 3.16 states that for every positive integer m the value semigroup of a generic parametrization with ramification profile $(2m, 2m+2, 2m+4)$ is

$$\begin{aligned} &\langle 2m,\ 2m+2,\ 2m+4,\ 4m+5,\ (m+2)m+1 \rangle && \text{if } m \text{ is even,} \\ &\langle 2m,\ 2m+2,\ 2m+4,\ 4m+5,\ (m+3)m+1 \rangle && \text{if } m \text{ is odd.} \end{aligned}$$

These are the semigroups S^* of [CKV23, Theorem 2.9]. For $m = 3$ the statement reads $S_{\mathrm{gen}} = \langle 6, 8, 10, 17, 19 \rangle$.

Proposition 7.1. *Let* $f_1 = t^6(1 + a_{1,1}t + O(t^2))$, $f_2 = t^8(1 + a_{2,1}t + O(t^2))$ *and* $f_3 = t^{10}(1 + a_{3,1}t + O(t^2))$. *Then*

$$f_3^2 - f_1^2 f_2 = (2a_{3,1} - 2a_{1,1} - a_{2,1})\, t^{21} + O(t^{22}).$$

Hence 21 *is a value for generic coefficients, while* $21 \notin \langle 6, 8, 10, 17, 19 \rangle$. *In particular the generic semigroup for* $(6, 8, 10)$ *is not* $\langle 6, 8, 10, 17, 19 \rangle$.

Proof. The expansion follows from $f_3^2 = t^{20}(1 + 2a_{3,1}t + O(t^2))$ and $f_1^2 f_2 = t^{20}(1 + (2a_{1,1} + a_{2,1})t + O(t^2))$. The coefficient $2a_{3,1} - 2a_{1,1} - a_{2,1}$ is a nonzero polynomial. A representation of the odd number 21 in $\langle 6, 8, 10, 17, 19 \rangle$ would use an odd number of odd generators; one odd generator leaves 4 or 2, which are not in $\langle 6, 8, 10 \rangle$, and three odd generators already exceed 21. □

The tie at weight 20 involves the monomials f_3^2 and $f_1^2 f_2$ of different total degrees. In the proof of [CLM22, Theorem 3.16] (pp. 27–28), the case of even m is reduced to ties of two monomials at weights divisible by $4m+4$, and the case of odd m is said

to be analogous; the tie at weight 20 for $m = 3$ is not of this kind. By Example 1.2 the generic semigroup is $\langle 6, 8, 10, 17, 19, 21\rangle$, of genus 10, whereas $\langle 6, 8, 10, 17, 19\rangle$ has genus 11. Computing $G(2m, 2m+2, 2m+4)$ with Remark 5.1 for $1 \le m \le 15$, we found that it coincides with the semigroup of [CLM22, Theorem 3.16] exactly for $m \in \{1, 2, 4, 6\}$; Table 1 shows the cases $m = 3, 4, 5, 10$.

Theorem 3.16 of [CLM22] also states the genus as $\frac{1}{2}m^2 + 2m - 1$ for even m and $\frac{1}{2}m^2 + 2m - \frac{3}{2}$ for odd m. For $m = 3$ this gives 9, while $\langle 6, 8, 10, 17, 19\rangle$ has genus 11 and the generic semigroup has genus 10. For $m = 2, 4, 6$, where the generators agree, the formula gives $5, 15, 29$, and the generic genus is $6, 16, 30$.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

References

[AS69] K. Atkinson and A. Sharma, *A partial characterization of poised Hermite–Birkhoff interpolation problems*, SIAM J. Numer. Anal. **6** (1969), no. 2, 230–235.

[CKV23] E. Cotterill, N. Kaplan, and R. Vieira Costa, *Cusps in $\mathbb{C}^3$ with prescribed ramification*, preprint (2023), arXiv:2303.09303v2.

[CLM22] E. Cotterill, V. L. Lima, and R. V. Martins, *Severi dimensions for unicuspidal curves*, J. Algebra **597** (2022), 299–331, doi:10.1016/j.jalgebra.2021.11.028; theorem and page numbers as in arXiv:2006.09580v4.

[GMO26] E. Gorsky, M. Mazin, and A. Oblomkov, *Generic curves and non-coprime Catalan numbers*, Algebr. Geom. **13** (2026), no. 1, 25–57, doi:10.14231/AG-2026-002.

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